

Calibration results

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Normalized Residuals

Reprojection error (cam0): mean 0.1858209418800756, median 0.1667109136857323, std: 0.1140893346995224
Gyroscope error (imu0): mean 0.39075009161961577, median 0.3517703608462338, std: 0.24230882970847717
Accelerometer error (imu0): mean 0.747545649476568, median 0.6263156704126951, std: 0.5078389764531962

Residuals

Reprojection error (cam0) [px]: mean 0.1858209418800756, median 0.16671091368573232, std: 0.1140893346995224
Gyroscope error (imu0) [rad/s]: mean 0.0003007318155343248, median 0.0002707319627999848, std: 0.00018648741443972515
Accelerometer error (imu0) [m/s^2]: mean 0.01419700597438095, median 0.011894667999102788, std: 0.009644619011264175

Transformation (cam0):

T_ci: (imu0 to cam0):

```
[ [ 0.99975956 -0.02165021 0.00347594 0.04064337]
  [ 0.00341171 -0.00300288 -0.99998967 0.04257857]
  [ 0.02166043 0.9997611 -0.0029283 -0.01580789]
  [ 0. 0. 0. 1. ] ]
```

T_ic: (cam0 to imu0):

```
[ [ 0.99975956 0.00341171 0.02166043 -0.04043646]
  [-0.02165021 -0.00300288 0.9997611 0.01681191]
  [ 0.00347594 -0.99998967 -0.0029283 0.04239057]
  [ 0. 0. 0. 1. ] ]
```

timeshift cam0 to imu0: [s] ($t_{imu} = t_{cam} + \text{shift}$)

0.008346290578647407

Gravity vector in target coords: [m/s^2]

[-0.11259634 -9.79696407 0.41861678]

Calibration configuration

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cam0

Camera model: pinhole

Focal length: [605.6431395442901, 604.857587153714]

Principal point: [431.53130932256033, 238.96893044451534]

Distortion model: radtan

Distortion coefficients: [0.1075688915368883, -0.23684972414089894, -0.003872857813206929,
1.269926469583267e-06]

Type: aprilgrid

Tags:

Rows: 6

Cols: 6

Size: 0.035 [m]

Spacing 0.0105 [m]

IMU configuration

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IMU0:

Model: calibrated

Update rate: 100.0

Accelerometer:

Noise density: 0.0018991490331489059

Noise density (discrete): 0.018991490331489058

Random walk: 0.00016083566625242938

Gyroscope:

Noise density: 7.696269866190555e-05

Noise density (discrete): 0.0007696269866190555

Random walk: 3.0049216627512e-07

T_ib (imu0 to imu0)

[[1. 0. 0. 0.]

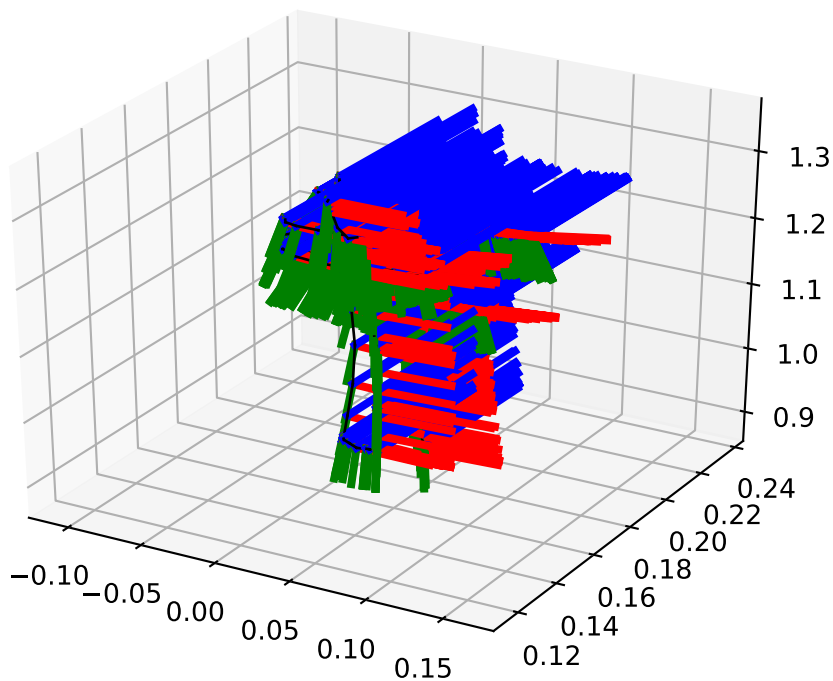
[0. 1. 0. 0.]

[0. 0. 1. 0.]

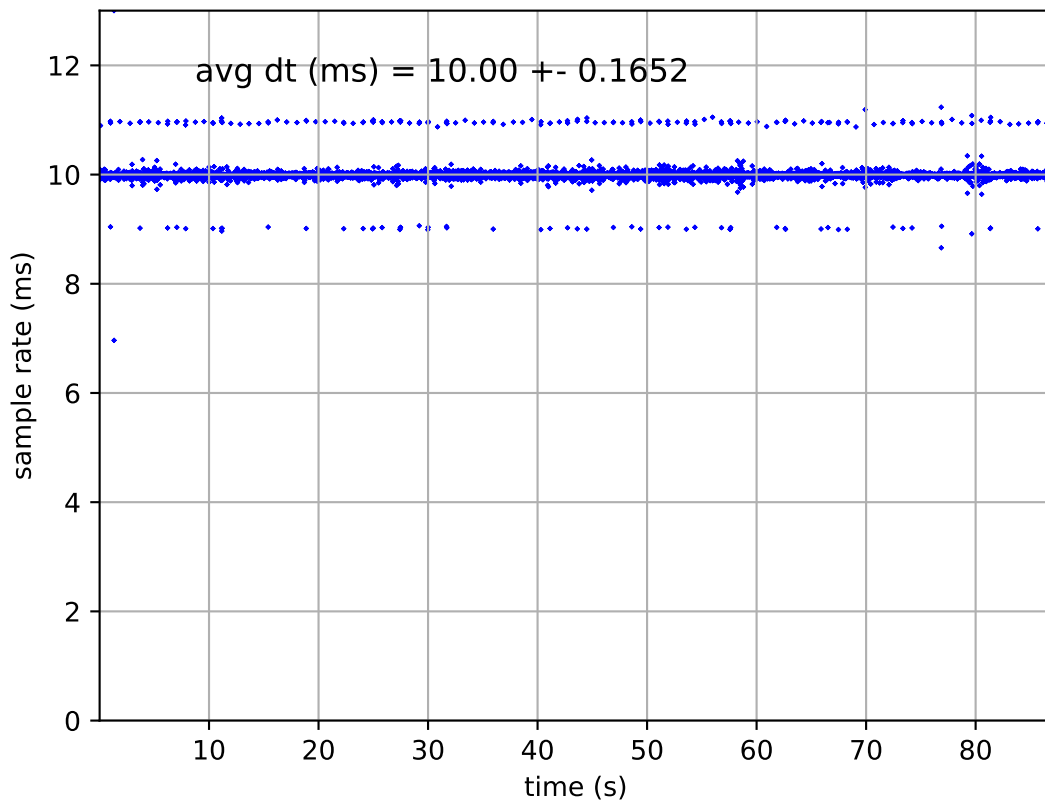
[0. 0. 0. 1.]]

time offset with respect to IMU0: 0.0 [s]

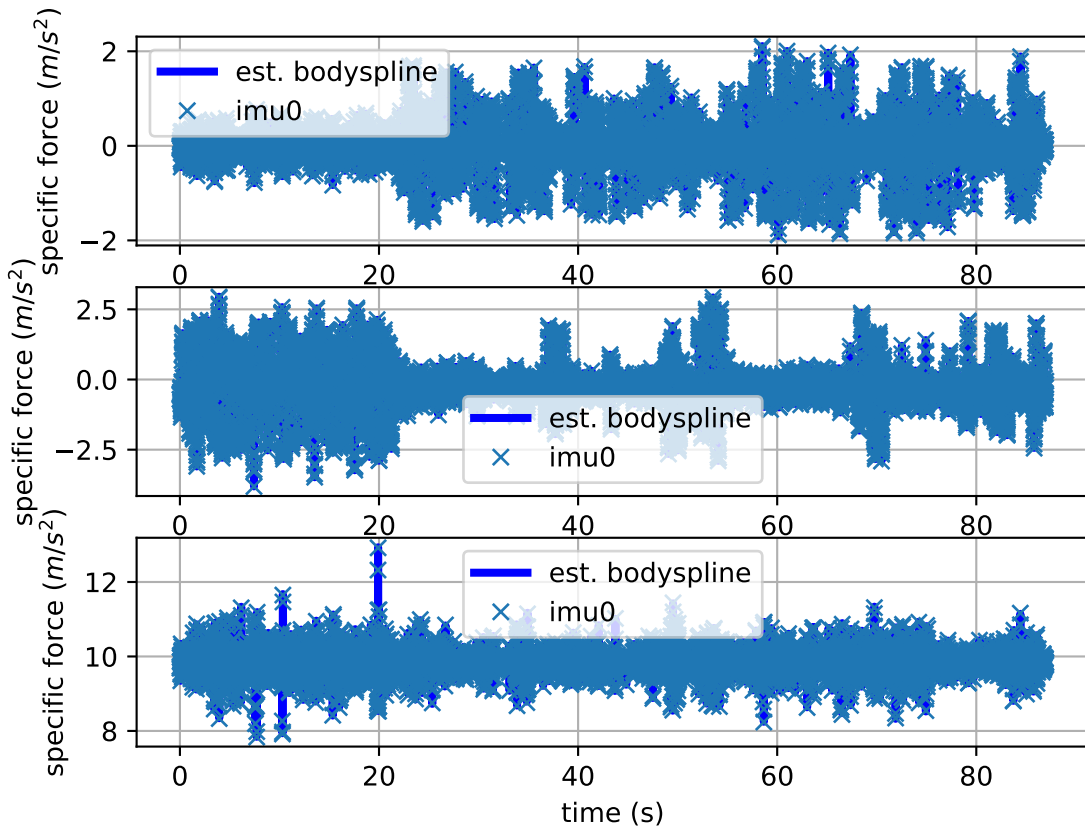
imu0: estimated poses



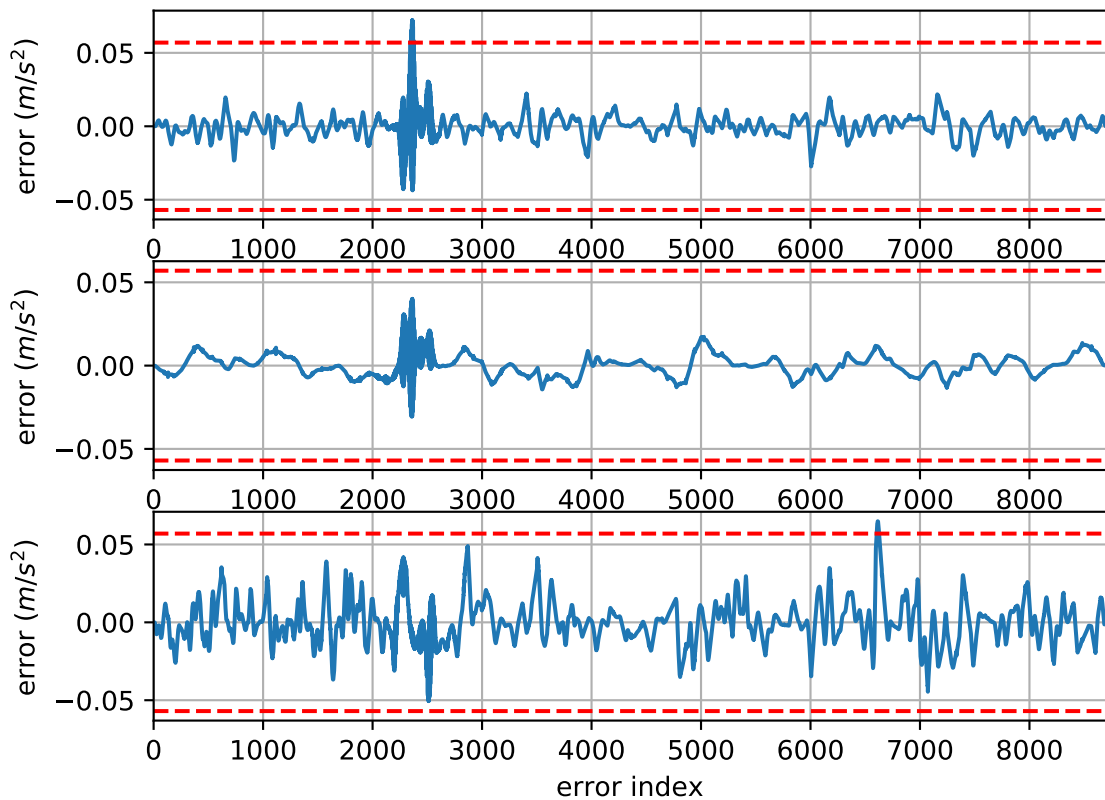
imu0: sample inertial rate



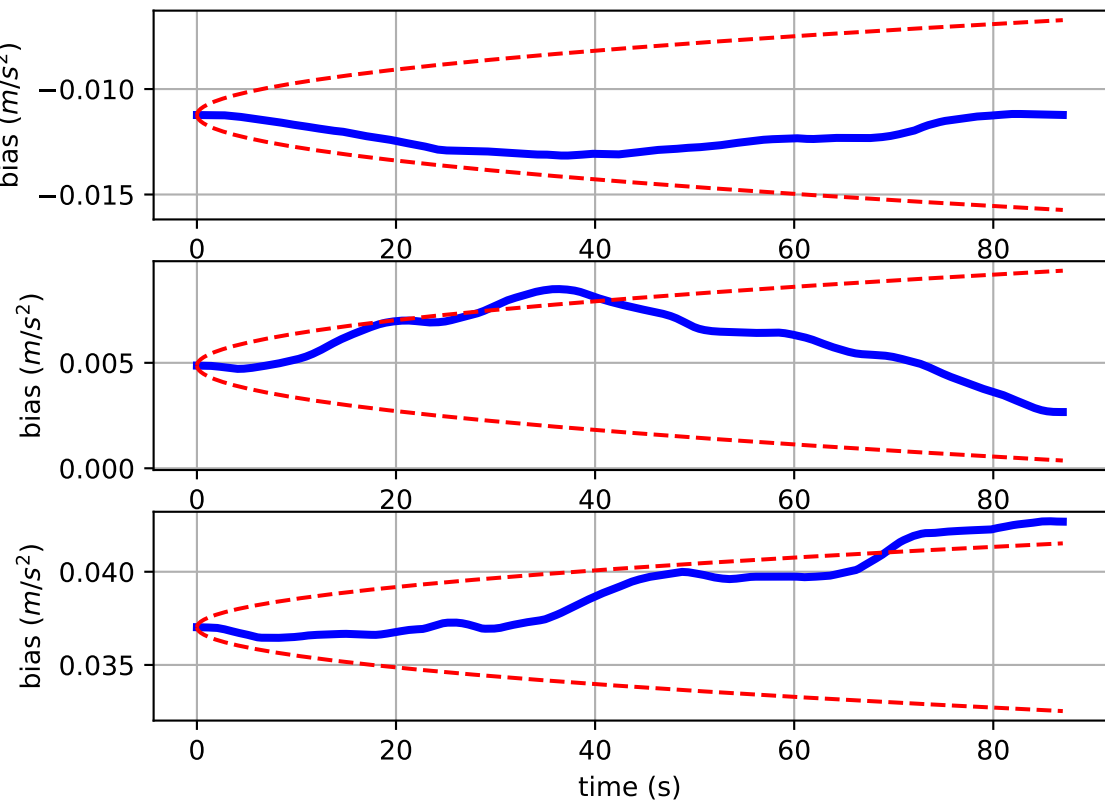
Comparison of predicted and measured specific force (imu0 frame)



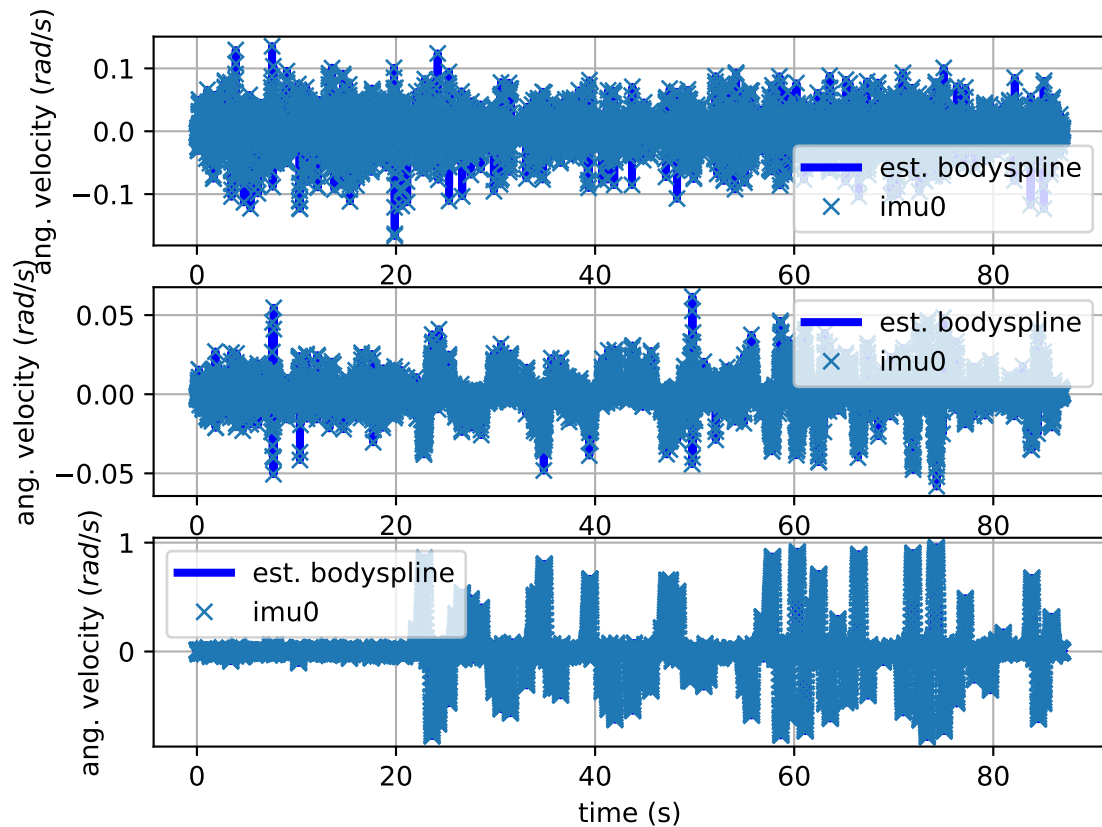
imu0: acceleration error



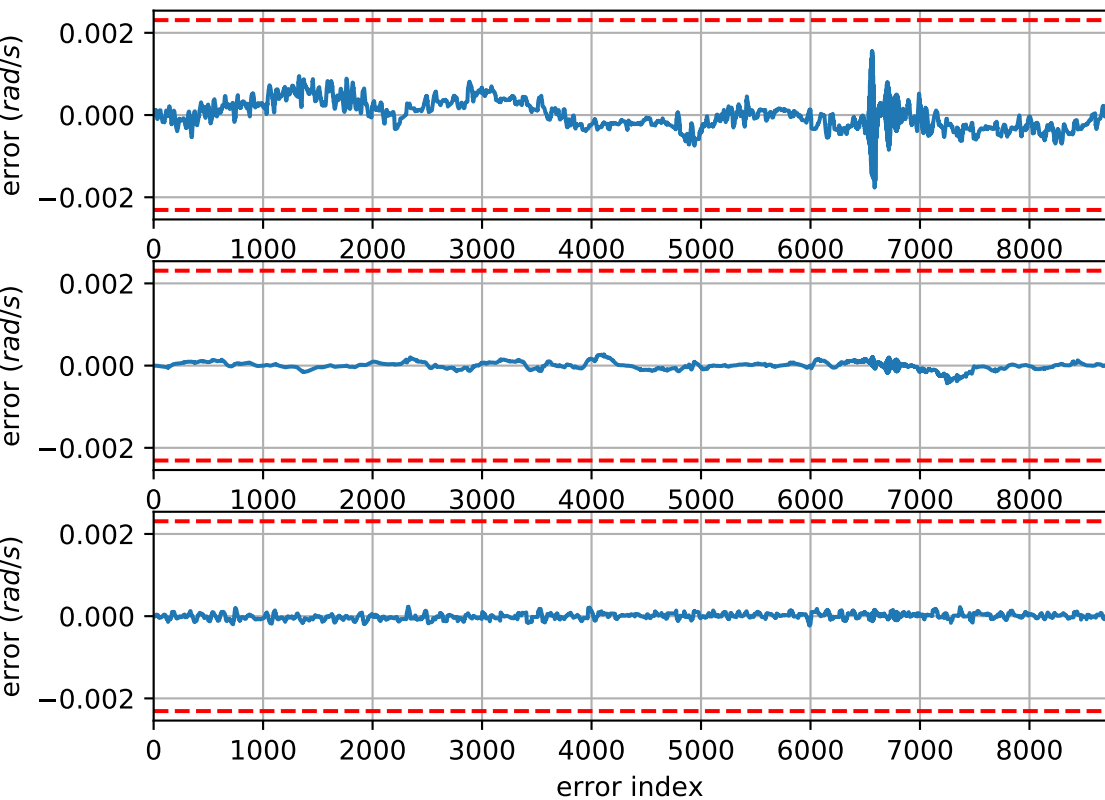
imu0: estimated accelerometer bias (imu frame)



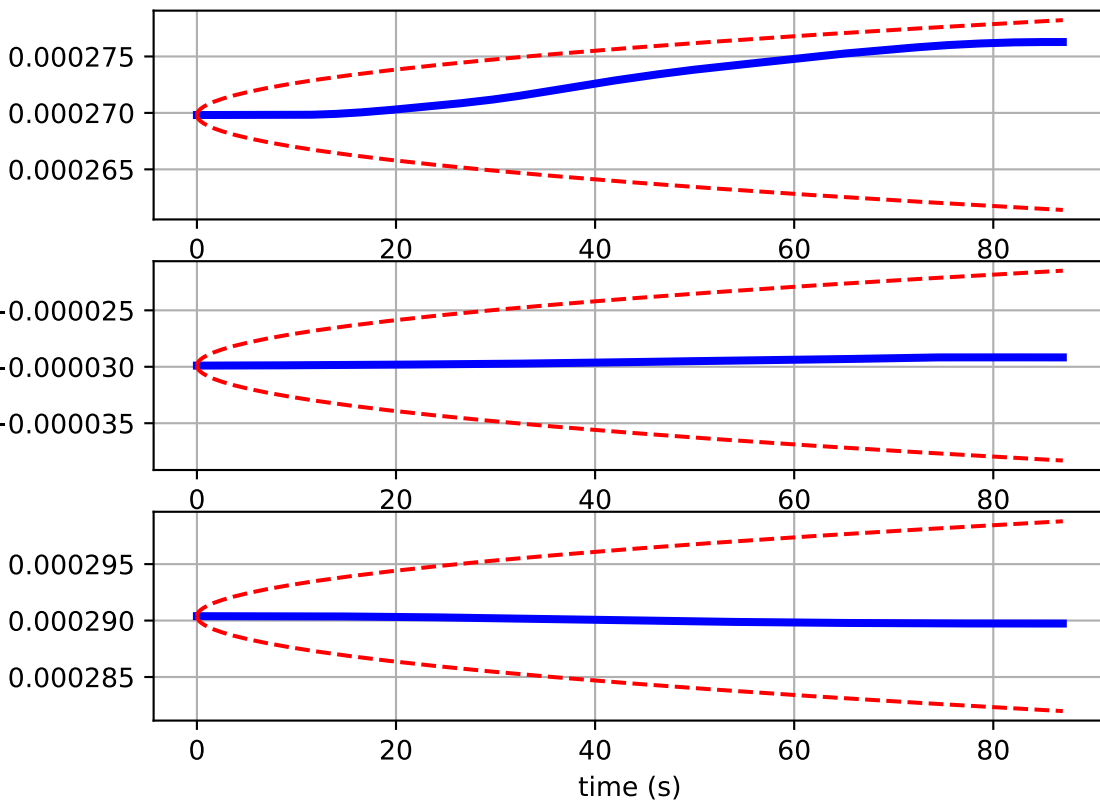
Comparison of predicted and measured angular velocities (body frame)



imu0: angular velocities error



imu0: estimated gyro bias (imu frame)



cam0: reprojection errors

